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REVIEW

Stromal vascular fraction: A regenerative reality? Part 1: Current concepts and review of the literature

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Summary Stromal Vascular Fraction (SVF) is a heterogeneous collection of cells contained within adipose tissue that is traditionally isolated using enzymes such as collagenase. With the removal of adipose cells, connective tissue and blood from lipoaspirate, comes the SVF, a mix including mesenchymal stem cells, endothelial precursor cells, T regulatory cells, macrophages, smooth muscle cells, pericytes and preadipocytes. In part 1 of our 2-part series, we review the literature with regards to the intensifying interest that has shifted toward this mixture of cells, particularly due to its component synergy and translational potential. Trials assessing the regenerative potential of cultured Adipose Derived Stem Cells (ADSCs) and SVF demonstrate that SVF is comparably effective in treating conditions ranging from radiation injuries, burn wounds and diabetes, amongst others. Aside from their use in chronic conditions, SVF enrichment of fat grafts has proven a major advance in maintaining fat graft volume and viability. Many SVF studies are currently in preclinical phases or are moving to human trials. Overall, regenerative cell therapy based on SVF is at an early investigative stage but its potential for clinical application is enormous.

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Introduction

Liposuction aspirates obtained from cosmetic surgery are a rich source of adipose-derived stem cells (ADSCs), a population of recent interest for its regenerative potential. Since the time they were first characterized by Zuk et al. in 2001, ADSCs have been extensively studied for their multipotency, paracrine effects and implications for regenerative medicine.¹ However, several studies have suggested that the regenerative properties of lipoaspirate are found not only in the ADSCs, but also in the mixed cell population of the lipoaspirate they are derived from, the stromal vascular fraction (SVF).^{2–9} This regenerative mix, which is traditionally isolated by enzymatic processing, contains multiple cell populations including mesenchymal stem cells (MSCs), endothelial progenitor cells (EPCs), immune cells, smooth muscle cells, pericytes and other stromal components.² While its cellular proportions may vary significantly based on processing and protocol, SVF has the regenerative potential to be used in therapies for numerous diseases such as multiple sclerosis² and diabetes.^{10,11} In this review, we discuss the growing research involving SVF and its numerous applications in animal and clinical trials.

Literature search

A search of the literature was conducted using PubMed, Google Scholar and Ovid using the terms "fat," "graft," "adipose," "ADSC," "regeneration," and "stromal vascular fraction." The literature was analyzed by searching recent conference proceedings, abstracts were assessed and relevant articles were included for full text review. References from the resulting citations for crosschecked for completeness. In total, 51 articles were included for review.

SVF as a regenerative modality (Table 1)

Fat grafting

The increasing use of autologous lipotransfer, or fat grafting, in plastic and reconstructive surgery has warranted an exploration of novel ways to improve clinical outcomes.^{12,13} As it currently stands, clinicians are challenged by significant variability in fat graft retention. Reported resorption rates range from 25 to 80%.¹⁴ Much of the volume loss is attributed to the tendency of mature adipocytes to undergo cell death after injection into the recipient site.¹⁵ Fat cell survival appears to be dependent upon its location within the graft: peripherally, adipocytes often survive while centrally, they necrose. The peripheral zone of the graft is the area of regeneration, where ADSCs stimulate the replacement and survival of adipocytes.¹⁵ Centrally, cell death occurs most prominently as a result of ischemia and poor tissue oxygenation.^{14–16}

In 2006, Matsumoto et al. introduced the concept of cell-assisted lipotransfer (CAL) in mice as a means of improving adipose tissue graft survival.¹⁷ The freshly isolated SVF (using collagenase and centrifugation) from one half of the lipoaspirate is combined with the prepared fat graft from the other half, which acts as a scaffold for the embedded SVF.¹⁷ The fat graft supplemented with SVF resulted in 35% greater graft retention and exhibited more prominent microvasculature than the non-SVF fat, suggesting its clinical potential.¹⁷ The use of CAL has since extended beyond animal models and exhibited efficacy in soft tissue augmentation in human trials. Yoshimura et al. applied CAL to cosmetic breast augmentation and observed 100–200 ml increases in breast volume after injecting a mean amount of 270 ml. CAL resulted in a natural texture and contour of the breasts, and most patients were satisfied with the outcome.³ These results appear to suggest that CAL is safe and effective when compared to traditional lipotransfer.

The usefulness of CAL has since been substantiated by researchers beyond Yoshimura's group. Li et al. revealed the clinical efficacy of SVF-assisted lipotransfer in the setting of cosmetic facial contouring, demonstrating markedly higher fat survival with the cell-supplemented grafts.⁴ Tanikawa et al. used fat grafts supplemented with adipose stromal vascular cells to successfully augment the soft tissue of patients with craniofacial microsomia. Their study showed significant enhancements in adipocyte viability and surviving fat volume in fat grafts enriched with ADSCs.¹⁸

Despite the early positive findings with regards to the safety and efficacy of SVF CAL procedures in promoting fat graft retention, more recent evidence is less clear. Peltoniemi et al. conducted an investigation on 18 women undergoing breast augmentation and discovered that a modified form of CAL (using adipose-derived regenerative cells, ADRCs – a proprietary mix of cells similar to SVF) generated by the Celution® system by Cytosol Therapeutics (San Diego, CA, USA) did not exhibit improvements over fat prepared after water-assisted liposuction.¹⁹ The authors suggest that the ADRCs may have lost their regenerative potential due to excessive time spent in enzymatic digestion and preparation. Meanwhile, the processing technique associated with water-assisted liposuction device advantageously washed and filtered the lipoaspirate, facilitating fat graft preparation and speeding up the processing time considerably in comparison to the Celution device.¹⁹ Further studies are needed to elucidate the underlying cause of Peltoniemi's results as successful regenerative enrichment of grafts with the Celution® system has been reported in other studies.^{20,21}

One topic of controversy includes the use of CAL in the setting of breast reconstruction after cancer excision. Chatterjee et al. recently demonstrated ex-vivo that SVF is a powerful adjunct for the expansion of epithelial progenitors in breast tissue, and that this effect is far greater in adipose tissue adjacent to breast tumors when compared to contralateral breast tissue from the same subject or adipose tissue from healthy donors.²² While these findings support the use of SVF in post-mastectomy reconstruction as an alternative or adjunct to achieve a desired aesthetic result, caution must be observed as previous studies have implicated ADSCs found in SVF as potential sources for breast cancer cell proliferation, migration and metastasis.^{23–25}

In the absence of cancer, however, recent clinical studies demonstrate the therapeutic potential of CAL for use in humans. For instance, Kõlle et al. showed level one evidence that patients treated with ADSC-enriched fat grafts retained significantly higher residual fat volumes compared to the control group.¹⁴ And in a small clinical trial of six patients, Charles-de-Sa et al. compared the use of SVF and expanded ADSCs as antiaging treatment of facial skin. From each patient, the researchers processed abdominal fat via two different protocols to obtain SVF and expanded ADSCs. The samples were injected with fat grafts subdermally into the right and left periauricular areas, respectively, of the same patient. Interestingly, no significant difference was found between the SVF and ADSC treatments histologically or clinically. Both the SVF and ADSC groups demonstrated reversal of histologic findings of aging, including decreased elastosis, new elastic fiber

formation, vascular proliferation, and collagen remodeling. Clinically, facial skin appeared more hydrated and rejuvenated with both treatments. The authors note that these are promising results for cosmetic surgery because SVF is easier and simpler to obtain than ADSCs.²⁶

While the studies cited above successfully utilize SVF or cells derived from the SVF as their enrichment modality for fat grafting, the method of isolation must be considered. In the U.S., for instance, the use of collagenase and cell culture techniques for the isolation of SVF and ADSCs precludes their immediate use for direct translational applications in humans.

Nanofat

By convention, SVF is extracted from adipose tissue after digestion with collagenase for 30 min to 1 h.^{27,28} Despite the commonality of such a procedure, the US Food and Drug Administration (FDA) regulates the on-label usage of collagenase-based products with tissue or cells. Since harvested adipose tissue meets the classification of a human tissue or cell-based product, the FDA restricts the on-label use of collagenase on the basis that it exceeds 'minimal manipulation' and combines tissue with an extraneous article. More specifically, the FDA recently released draft guidelines declaring any adipose tissue, even if intended for autologous use, as more than 'minimally manipulated' if it is not "composed of clusters of adipocytes and other cells surrounded by a reticular fiber network and interspersed with small blood vessels, divided into lobes and lobules by connective tissue septa".²⁹ Consequently, SVF is subject to a separate set of regulations under Section 351 of the Public Health System Act for biological products.^{12,30,31} Furthermore, in consideration of the time practicalities in plastic and reconstructive procedures, the time allotted to collagenase digestion poses yet another inconvenience.

These barriers to entry necessitate a more efficient methodology to isolate an SVF-like pellet without collagenase. Tonnard et al. recently described a potential clinical alternative which they termed nanofat. In nanofat grafting, the lipoaspirate is emulsified by repeatedly shifting the fat between two connected syringes. After filtration and disposal of the fat and stroma, the vestigial effluent is injected into the patient to correct superficial rhytides, scars and eyelid discoloration. In essence, nanofat grafting employs intense mechanical shearing to rupture adipocytes, leaving a nanofat sample that is rich in stem cell content. Hence, despite its name, nanofat grafting more closely resembles a transfer of regenerative cells than a transfer of fat.^{32,33} Nanofat grafting signifies the potential to shift from enzyme-based reagents to mechanical processes for SVF isolation where volumetric replacement is not required. However, the exact constituents and regenerative potential of this mix is yet to be well-defined.

SVF & nerve regeneration potential

Several studies have suggested that both ADSCs and SVF enhance peripheral nerve regeneration. The effectiveness of cultured ADSCs has been well demonstrated in cell implanted fibrin conduit studies. Di Summa et al. differentiated adipose and bone marrow derived stem cells (BMSCs) to a Schwann cell (SC) phenotype and transplanted the cells onto transected sciatic nerve gaps.³⁴ Transected

nerves transplanted with either stem cell population showed greater axonal regeneration distances compared to empty conduits. Moreover, transplantation with differentiated ADSCs showed no significant difference in regeneration distance compared to transplantation with differentiated BMSC.³⁴ This suggests that nerve regenerative potential of ADSCs are similar to their bone marrow derived counterparts, a cell population which has already been successfully used to stimulate nerve regeneration.³⁵

While ADSCs have demonstrated strong results in murine models of nerve regeneration, the use of a heterogeneous SVF cell population also shows promise. Mohammadi et al. demonstrated that uncultured SVF significantly enhanced nerve regeneration *in vivo* in a sciatic nerve transection model.³⁶ The animals treated with uncultured SVF showed increased nerve function, functional recovery time, and muscle mass compared with the control group. Immunohistochemistry studies of the transection nerve also showed that SVF grafting increased the presence of SC-like cells compared to normal repair. Subsequently, there was significantly increased axon diameter, nerve fiber, and myelin sheath thickness.³⁶ The promising results of this study suggest that SVF may enhance nerve regeneration by providing extracellular matrix proteins and factors that supports SC proliferation and axonal growth.

Although both expanded ADSCs and uncultured SVF have been demonstrated to enhance nerve regeneration, the use of whole adipose tissue hinders nerve regeneration. Papalia et al. demonstrated that nerve regeneration and functional recovery were significantly worse than control when transected nerve conduits were filled with adipose tissue that was not centrifuged or chemically altered.³⁷ These results suggest that SVF must first be isolated from adipose tissue to significantly enhance peripheral nerve regeneration.

Multiple sclerosis

Multiple sclerosis (MS) is an autoimmune disease that causes demyelination of central nervous system neurons. Experimental autoimmune encephalitis (EAE), an experimentally induced demyelinating disease, is often used as an animal model of chronic multiple sclerosis.³⁸ Animal studies examining the effects of SVF and ADSCs on EAE models suggest that SVF may be clinically potent against MS.

Semon et al. compared the efficiency of expanded ADSCs and SVF in treating chronic EAE and found that both reduce the demyelination and breakdown products commonly seen in EAE.³⁹ This demonstrates that ADSCs and SVF both have neuroprotective properties that reduce the pathological features of EAE. However, SVF injection is more effective than ADSC injections at delaying EAE onset and reducing disease scores.

Burn injuries

Skin grafts are currently the main method of treating burn wounds. Skin grafts however, have several drawbacks, including the availability of donor areas, rejection by the host immune system, and poor retention due to low vascularity, hematoma, and infection. Increasing the speed with which the graft is vascularized will accelerate the rate of wound healing. Using the combination of SVF cells rather than isolated ADSCs may demonstrate even greater benefits and healing potential due to the inclusion of other cell types and growth factors present in SVF.^{6,10}

SVF cells can improve burn healing when grafted on the wound bed by stimulating host cells, and also by secreting growth factors and extracellular matrices due to their autologous nature.¹⁰ Atalay et al. have shown in rats, that administering SVF isolated from the adipose tissue through intradermal injections can accelerate and improve the healing of deep partial thickness burns. The SVF has been found to increase vascularization and fibroblastic activity in the treated area, and also reduce inflammation and increase cellular proliferation when compared to a control rat group.⁶

Diabetes

The regenerative capacity of SVF extends to serious complications of diabetes mellitus, namely the diabetic foot ulcer (DFU), nephropathy and retinopathy.^{10,11} As one of the most frequent complications of diabetes, the DFU is a condition rooted in ischemia and neuropathy. As it progresses, the DFU becomes susceptible to debilitating infections.⁴⁰ An *in vitro* study by Han et al. observed that SVF significantly increased the proliferation (28%) of DFU fibroblasts, a major cell component involved in wound healing. Furthermore, the study also noted increased collagen synthesis (44%) by the fibroblasts, a process that is important for the synthesis of an extracellular matrix. Han et al. followed up the *in vitro* study with a clinical study and demonstrated that SVF accelerated wound healing when the SVF cells were dispersed onto the debrided wound (autograft). The authors noted complete wound healing in 100% of the SVF-treated patients, in contrast to 62% for the control group. Han et al. suggest SVF was able to secrete growth factors and ECM to enhance the regenerative microenvironment.^{10,41}

A recent study by Eto et al. characterized various adipose tissue macrophage (ATM) populations in the stromal vascular fraction noting that over 90% of ATMs exist in the M2 state in SVF.¹⁵ The anti-inflammatory properties of the M2 phenotype likely also contribute to the potency of SVF therapy for DFUs.

Retinopathy is cited as the most common diabetic complication, one characterized by dysfunction and degeneration in the vascular network of the retina.¹¹ A review by Jarajapu et al. suggests that SVF cell types such as the EPCs may facilitate retina repair, a hypothesis broached through experiments on pre-clinical mouse models. In a more recent study, diabetic athymic rats were injected with ADSCs and a decrease in vascular leakage and apoptosis was observed; the authors suggest that ADSC crosstalk with retinal endothelial cells promotes vascular formation and hinders neurodegeneration.⁷ Additionally, MSCs have demonstrated pre-clinical capacity to heal kidney injuries in rats with diabetic nephropathy.⁴²

Radiation

Radiation therapy commonly causes altered blood vessel flow with resultant ischemia in treated tissue. The altered vasculature and blood supply caused by radiation often results in tissue desquamation, erythema, edema, fibrosis, and necrosis.⁴³ Rigotti et al. studied the effects of heterogeneous lipoaspirate injection as a treatment option for radiotherapy patients with severe or irreversible functional damage.⁴⁴ Injection of the lipoaspirate into radio-damaged tissue resulted in enhanced neovascularization with

increased tissue hydration, perfusion, and oxygenation. The neovascularization induced by SVF supplanted the poor vascularization and fibrosis that resulted from radiation therapy. As a result, radio-damaged and fibrosed tissue underwent transformative changes after SVF injection. Clinically, these patients showed dramatic systemic improvements with decreases in disease scores after a mean follow up of 30 months.⁴⁴

Crohn's disease

Along with breast reconstruction, fistula repair procedures for Crohn's Disease are among the most frequent applications of SVF cells for clinical therapy.⁴⁵ Crohn's Disease is an inflammatory bowel disorder characterized by abnormal inflammation of the gastrointestinal tract. The result is a host of symptoms that range from abdominal pain to bloody diarrhea to severe weight loss.⁴⁶ While the precise cause of Crohn's Disease remains indeterminate, evidence suggests that environmental agents and genetic abnormalities are the main contributors. These factors can subsequently lead to the dysfunction of mucosal barriers, immunomodulation and bacterial clearance.⁴⁷

The potency of SVF cell components for the treatment of fistulas from Crohn's Disease has been documented in *in vitro* and in clinical studies. *In vitro* studies demonstrate that MSCs have the capacity to inhibit T-cell alloreactivity, a major characteristic of Crohn's Disease.⁴⁸ Such an effect is likely an extension of the MSC's ability to immunoregulate its immediate microenvironment, a property found in other cells of the SVF including M2 macrophages. In human trials, MSCs have been injected systemically and locally into the Crohn's Disease patients. Outcomes of MSC-based therapy have demonstrated short-term safety and efficacy, but further clinical tests are needed to elucidate optimal dosage and administration of MSCs.⁴⁸ Thus, although there is no definitive cure for Crohn's Disease, cell therapy with SVF components has exhibited early promise.

Bone

ADSCs have been shown to be highly successful in inducing bone regeneration and healing *in vivo* in animal studies⁸ as well as in human trials.⁴⁹ In one such study, cranioplasty procedures were performed in four human patients using ADSCs seeded in a scaffold of beta-tricalcium phosphate (β -TCP) granules. Along with a satisfactory outcome in ossification, no clinically relevant complications were observed.⁴⁹ In another study, a 10 cm anterior mandibular defect was replaced using a scaffold of β -TCP, recombinant human bone morphogenetic protein-2 (BMP-2) and ADSCs. The implant was left to mature over a period of ten months, at which time there was the sufficient ossification of the graft for successful transplantation to the donor site.⁵⁰ Aside from inducing bone formation, ADSCs also appear to accelerate the healing of bone when directly injected into a bone defect or when administered systemically into swine.⁵¹

Given the success of ADSC interventions, research has shifted towards the application of SVF in bone regeneration. Mehrkens et al. demonstrated the osteogenic potential of SVF using recombinant human BMP-2 to induce SVF cells to differentiate into osteogenic cell lineages to generate new bone tissue.⁵² Additionally, a study by Jurgens et al.

compared the degree to which osteochondral defects created in the knees of goats could be repaired using either freshly isolated SVF or cultured ADSCs seeded onto collagen scaffolds. The results showed similar bone regeneration in the scaffolds seeded with SVF cells versus cultured ADSCs. Both SVF and ADSC scaffold groups demonstrated greater degrees of regeneration compared to unseeded control scaffolds, with no noted adverse effects.⁵³

The findings of Mehrkens et al. and Jurgens et al. suggest that ADSCs and SVF may have the potential to be grafted into injured bone to facilitate bone repair and regeneration. This has led to an increased interest into the osteogenic potential of SVF and its application into orthopedic procedures. At the time of writing, several clinical trials have begun investigating the therapeutic applications of SVF and ADSCs in human bone repair.

Cardiac

Many studies have attempted to apply stem cells, such as ADSCs and BMSCs, into treatments for various cardiac conditions. The therapeutic value of transplanting ADSCs to treat myocardial infarction is limited. When ADSCs alone are implanted directly into heart muscle, poor long term cell engraftment and survival have been reported.⁵⁴ Given its heterogeneous nature with multiple progenitor types, SVF transplantation may be as or more effective than ADSCs treatment alone. Van Dijk et al. intravenously injected rats in an induced acute myocardial infarction (AMI) model with either uncultured rat SVF cells or cultured rat ADSCs. A significant reduction in the infarct size was observed in both treatment groups. The rats treated with SVF showed no adverse effects, whereas of the rats treated with ADSCs, three exhibited shortness of breath and one died during injection. These results suggest that uncultured SVF may have the same regenerative potential as cultured ADSCs, possibly with fewer side effects.⁵

One notable clinical application of SVF was recently reported in the American Journal of Medicine. In this case, a 73 year old male with refractory angina pectoris was injected with SVF cells both intramyocardially to the left apex as well as intravenously. The SVF cells proved effective in repairing the damaged myocardium and cardiac function was partially restored, demonstrated by an ejection fraction that improved from 20% to 35% in just 7 days. SVF's ability to repair damaged cardiac tissue in this case suggests its cardiomyogenic potential.⁵⁵

Discussion

In recent years, there has been much discussion regarding ADSCs as a form of cell-based therapy in regenerative medicine. As an example, Gimble et al. reviewed the capacity for ADSCs to differentiate along multiple lineages and described its potential benefits for cardiac diseases.⁵⁶ However, due to extensive culturing periods and FDA regulations, the limited clinical translatability of cultured ADSCs poses significant challenges for procedures that necessitate more efficient solutions. Thus, scientists and clinicians have begun to explore the regenerative capabilities of the SVF, a more heterogeneous and synergistic mixture of cells. Unlike ADSCs, SVF represents a less

Table 1 SVF applications.

Therapeutic target	Author	Model	Findings
Peripheral nerve regeneration	Mohammadi et al., 2014 ³⁶	Murine in vivo (n = 18)	SVF significantly increased nerve function, functional recovery time, and muscle mass compared to control in a nerve transection model.
Multiple sclerosis	Semon et al., 2013 ³⁹	Murine in vivo (n = 12)	SVF is comparable to ADSC in reducing demyelination in experimental autoimmune encephalitis. SVF is more effective than ADSCs at delaying disease onset, reducing disease scores, and decreasing inflammatory IL-12.
Burn wounds	Atalay et al., 2014 ⁶	Murine in vivo (n = 10)	SVF injection accelerates and enhances healing of deep partial thickness burns compared to control. SVF increases vascularization and proliferation, while reducing inflammation compared to control.
Retinopathy	Rajashekhar et al., 2014 ⁷	Murine in vivo (n = 8)	ADSCs decreased vascular leakage and apoptosis, increased crosstalk to promote vascular formation, and hindered neurodegeneration compared to control.
Cardiac	Premaratne et al., 2011 ⁹	Murine in vivo (n = 12)	SVF transplanted into ischemic myocardium enhanced vascular density, improved cardiac function, and reduced fibrotic tissue compared to control.
Lipotransfer	Van Dijk et al., 2011 ⁵	(n = 6)	
	Yoshimura et al., 2006 ⁵⁷	Human in vivo (n = 28)	Cell assisted lipotransfer (CAL) with SVF increased breast volume and resulted in natural texture and contour during cosmetic breast augmentation.
	Yoshimura et al., 2008 ³	(n = 40)	
	Yoshimura et al., 2010 [60]	(n = 188)	
	Li et al., 2013 ⁴	Human in vivo (n = 26)	SVF-assisted lipotransfer demonstrated higher fat survival compared to control for cosmetic facial contouring.
	Tanikawa et al., 2013 ¹⁸	Human in vivo (n = 7)	SVF significantly enhanced adipocyte viability and surviving fat volume compared to control for patients with craniofacial microsomia.
	Kolle et al., 2013 ¹⁴	Human in vivo (n = 10)	ADSC-enriched fat grafts retained higher volume compared to the control group.
	Tonnard et al., 2013 ³²	Human in vivo (n = 67)	'Nanofat' was injected into patients to correct rhytides and discoloration of eyelids. It may have implications for a new form of CAL.
	Charles-de-Sa et al., 2015 ²⁶	Human in vivo (n = 6)	SVF modifies the dermal aging process and produces a skin rejuvenation effect when injected subdermally into periauricular areas. SVF injections produces comparable effects to ADSC injections.
Diabetic foot ulcer	Han et al., 2010 ⁴¹	Human in vivo (n = 28)	SVF was dispersed onto the wound and accelerated healing in diabetic foot ulcers (DFU) compared to control. SVF increased the proliferation of DFU fibroblasts.
Radiation	Rigotti et al., 2007 ⁴⁴	Human in vivo (n = 20)	SVF injection enhanced neovascularization, hydration, perfusion, and oxygenation in radio-damaged tissue compared to control.
Crohn's disease	Dalal et al., 2012 ⁴⁸	Human in vivo (n = 24)	MSC therapy demonstrated short-term safety and efficacy in Crohn's treatment compared to control.
Bone regeneration	Mehrkens et al., 2012 ⁵²	Murine in vivo (n = 7)	Recombinant human BMP-2 stimulates SVF to differentiate into committed osteogenic progenitors in vitro, which is then capable of forming ectopic bone tissue when implanted into control mice.
	Jurgens et al., 2013 ⁵³	Caprine in vivo (n = 8)	SVF scaffolds induced significant osteochondral regeneration compared to acellular scaffolds when implanted into caprine knee defects. SVF scaffolds produced similar regeneration effects as cultured ADSC scaffolds.

ADSC — Adipose-Derived Stem Cell; CAL — Cell Assisted Lipotransfer; DFU — Diabetic Foot Ulcer; IL-12 — Interleukin 12; MSC — Mesenchymal Stem Cell; SVF — Stromal Vascular Fraction.

Table 2 Overview of SVF clinical trials.

#	Study	Intervention	Status
1	Rheumatoid arthritis	SVF	Active, not recruiting
2	Osteoarthritis	SVF	Active, not recruiting
3	Idiopathic pulmonary fibrosis	ADSC/SVF	Recruiting
4	Critical Limb Ischemia	ADSC/SVF	Recruiting
5	Comparison between cytori and manual preparation of ADGC for obesity on SVF cell counts	SVF	Completed
6	Breast augmentation/micromastia	Fat graft vs. SVF	Recruiting
7	Immunophenotyping of SVF from ADSCs	SVF	Unknown
8	Systemic sclerosis	SVF	Completed
9	Extraction of SVF from fat tissue	SVF	Completed
10	SVF derived MSC in renal transplantation	SVF	Recruiting
11	Osteoarthritis of knee	SVF	Completed
12	Facial fat grafting for enhancement	SVF	Recruiting
13	Type 2 diabetes	ADSC/SVF	Unknown
14	Type 1 diabetes	ADSC/SVF	Unknown
15	Recto-vaginal fistula	SVF	Unknown
16	Enterocutaneous fistula	SVF	Unknown
17	Critical limb ischemia	SVF	Recruiting
18	Multiple sclerosis	ADSC/SVF	Recruiting
19	Osteoarthritis	ADSC/SVF	Recruiting
20	Rheumatoid arthritis	SVF	Recruiting
21	COPD	SVF	Not yet recruiting
22	Osteoarthritis of knee	ADSC/SVF	Active, not recruiting
23	COPD	ADSC/SVF	Recruiting
24	Orthopedic, neurologic, urogenic, cardiopulmonary conditions	SVF	Recruiting
25	Amputation	SVF	Recruiting
26	Chronic wounds — diabetic foot, venous ulcer, pressure ulcer	ADSC/SVF	Recruiting
27	Osteoarthritis of knee	ADSC/SVF	Completed
28	Osteoarthritis	SVF	Recruiting
29	Skin quality improvement in irradiated breasts	SVF	Not yet recruiting
30	Parkinson's disease	SVF	Recruiting

processed form of cell-based therapy that bypasses the weeks spent culturing and passaging cells. Although SVF has not been conclusively proven to be superior to ADSCs, it has certainly demonstrated significant promise and warrants further exploration.

From *in vitro* studies that elucidate the crosstalk mechanisms between the various cell types to animal studies that display *in vivo* effectiveness, SVF application is now at a stage where it is advancing toward human trials for a number of conditions. In its present state, SVF is commonly used in lipotransfer to improve volume retention and enhance the natural appearance of the breasts during aesthetic or cosmetic surgery. One of the earliest to employ CAL with SVF was the Yoshimura group, who, along with others, have since conducted multiple studies detailing the effectiveness of CAL in human patients.^{3,26,57,58}

SVF has been similarly applied to other disease models. In diabetic foot ulcers and radio-damaged tissue, for instance, the application of SVF in human patients improves the healing process when compared to the control groups.^{41,44} Other conditions, such as burn wounds, retinopathy and nerve regeneration, are still being investigated in animal models and *in vitro*.

SVF-based therapy is progressing quickly as an effective and safe treatment option in tissue engineering and

regenerative medicine. However, as we continue to explore the clinical potency of SVF for numerous diseases, further research is required to fully characterize SVF and investigate the basic mechanisms underlying SVF-based regeneration. Since SVF was first described, researchers have developed different protocols to isolate and characterize the heterogeneous mixture of cells. This has led to moderate variation in the exact composition and cell surface marker expression of SVF used in different studies. While it is difficult to standardize the exact constituents and cellular composition of SVF, it is likely that there is a minimum concentration of progenitor cells needed for SVF to have regenerative and therapeutic effects. In *Stromal Vascular Fraction — A Regenerative Reality? Part 2*, we will characterize SVF in greater detail and discuss its potential mechanisms of regeneration.

Trends and status of SVF clinical trials

There are a growing number of clinical studies being conducted on SVF and its therapeutic benefits. In a search for “stromal vascular fraction” on ClinicalTrials.gov there are currently 26 registered clinical studies that directly involve SVF and human participants to date (Table 2).⁵⁹ The

majority of the studies registered on the database are ongoing and still recruiting human participants for their trials. While three of the registered studies focused on the characterization and extraction of SVF, most focuses on the therapeutic benefits of applying SVF in several human diseases and medical procedures.

The growing number of clinical trials on SVF represents a shift from the study of cultured and homogenous ADSCs to a heterogeneous mixture of SVF cells. The clinical studies registered on ClinicalTrials.gov studying the application of heterogeneous SVF in a variety of different diseases and therapies, such as arthritis, diabetes mellitus, chronic wounds, breast cancer, and cosmetic surgery.⁵⁹ As discussed previously, SVF has already demonstrated numerous modulatory and regenerative benefits in similar disease models in animal studies and case studies.^{57,48} The ongoing clinical studies, most of which are in Phase 1 or Phase 2, seek to demonstrate the safety and efficacy of SVF therapy in humans. Given the promising results of the past SVF research, these current human clinical studies will demonstrate whether or not SVF has the potential to be used as treatments in humans.

Preparation of SVF and future considerations

Stromal vascular fraction is conventionally isolated from lipoaspirate by collagenase digestion and high-speed centrifugation. However, FDA regulations on collagenase results in off-label use that restricts widespread practice. Furthermore, it remains to be investigated whether collagenase interferes with cellular surface receptor proteins and stromal components that may be found to strengthen the regenerative potential of SVF. Tonnard's recent study on 'nanofat' has demonstrated that pure mechanical stresses can yield a mixture of cells or cell fragments that may be concentrated with progenitors and stem cells.³² This nanofat, with little to no viable fat cells, appears to have properties affecting skin pigmentation and texture. This mix presents yet another area for investigation. Elucidating the cellular composition of nanofat and comparable non-enzymatically obtained mixes may give further clues as to the ideal components of an SVF mix. Moving forward, our laboratory is interested in isolating a non-enzymatically prepared SVF mixture (NESVF) that is optimized for this purpose.

Conclusion

As a heterogeneous composite of mesenchymal stromal cells, endothelial cells, pericytes, and macrophages, the SVF has versatile applications in regenerative medicine. Its potential has been demonstrated in lipotransfer and various disease models, such as multiple sclerosis, diabetes and radiation damage. Presently, SVF studies are in the pre-clinical stage or are advancing toward human trials. Furthermore, from a regulatory standpoint, enzymatic digestion of lipoaspirate may prove a hindrance to SVF application in the operating room. Thus, as SVF emerges as the preferred form of cell-based therapy, further studies are needed to elucidate its characterization and optimize procedures for efficient isolation.

Conflict of interest statement

The authors declare no potential conflicts of interest with respect to research, authorship and publication of this article.

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